

Durban and Lesedi market sizing

National balance, issue tree and deep dives

South Africa petrol and diesel, Durban-Lesedi corridor

9 October 2026

Draft for review. Lever cases are proposals, not accepted inputs;
open exceptions mean outputs are provisional.

Photo source: Royal Vopak | Terminal Durban | vopak.com



National petrol and diesel balance: what makes up the ~20B litres

Year	Domestic production	Gross imports	Exports	Sales	Supply less sales
2019 Energy balance, customs, sales	19.5B L Petrol 10.4 · diesel 9.1 observed · DR01	7.1B L + Petrol 1.5 · diesel 5.6 customs · DR01	2.9B L - Petrol 1.2 · diesel 1.7 customs · DR01	23.7B L - Petrol 10.8 · diesel 12.9 observed · DR01	0.0B L = Petrol 0.0 · diesel 0.0 Stocks + stat. difference
2021 Energy balance, customs, sales	11.7B L Petrol 6.4 · diesel 5.3 observed · DR01	13.8B L + Petrol 4.0 · diesel 9.8 customs · DR01	1.9B L - Petrol 1.1 · diesel 0.9 customs · DR01	22.2B L - Petrol 9.3 · diesel 12.9 observed · DR01	+1.2B L = Petrol 0.0 · diesel 1.2 Stocks + stat. difference
2023 Customs and sales	Not published No balance after 2021 Pending · DR08 operators	17.4B L + Petrol 4.5 · diesel 12.9 customs · DR01	1.9B L - Petrol 1.0 · diesel 0.9 customs · DR01	21.9B L - Petrol 9.0 · diesel 12.9 observed · DR01	Cannot close = Needs production Pending · production, stocks

Production fell as imports rose

Production 19.5B L (2019) to 11.7B L (2021)
Imports 7.1B L (2019) to 17.4B L (2023)

DR01 · energy balance, customs

The balance closed to 2021

Supply within 0.8B L of sales, 2014-2021
Exception: diesel 2021, +1.2B L

DR01 · calculated

It cannot close after 2021

No production published after 2021
Stock change never published

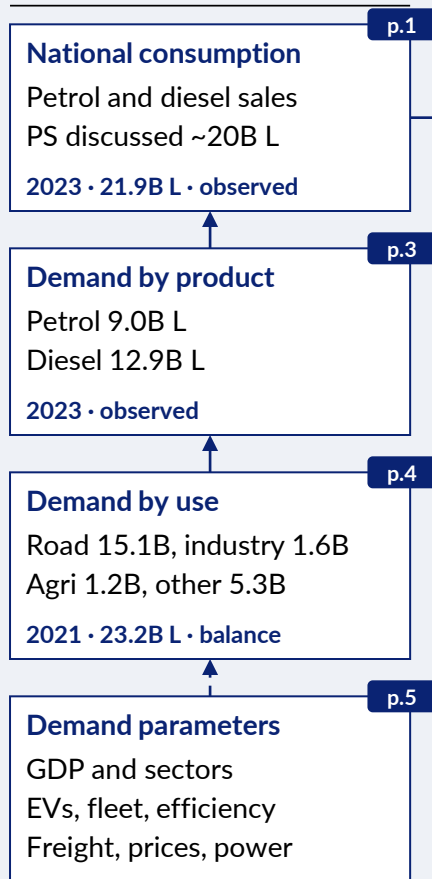
Pending · DR08 operators, DR01 stocks

Source: Manish DR01 (8 Oct): department energy balance (production to 2021), SARS customs (trade), department sales. Supply less sales = stock change + statistical difference, not production. FIASA and JODI not used. Petrol and diesel only.

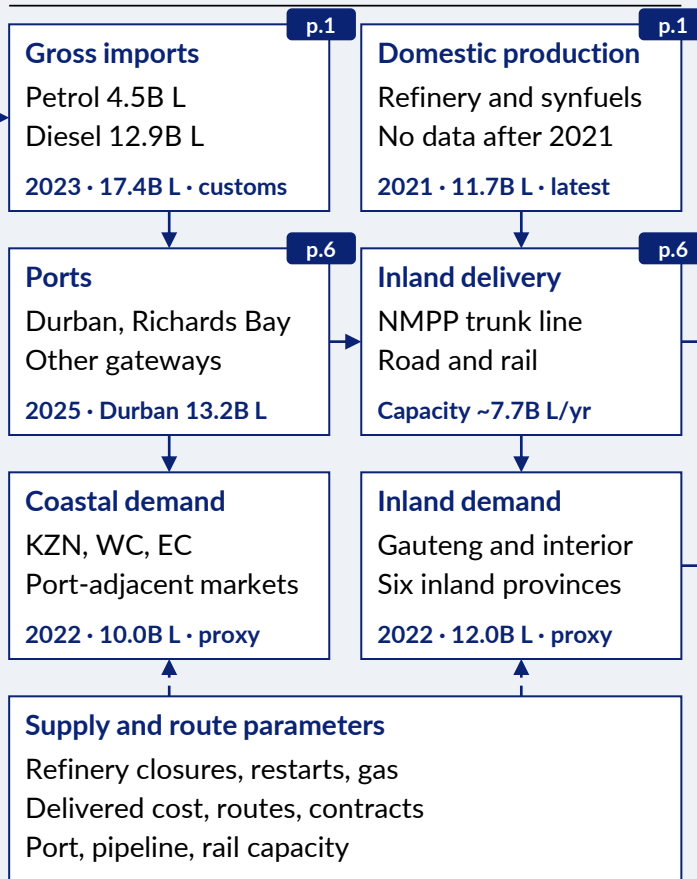
Durban and Lesedi market sizing and investment issue tree

p.# Deep dive on that page

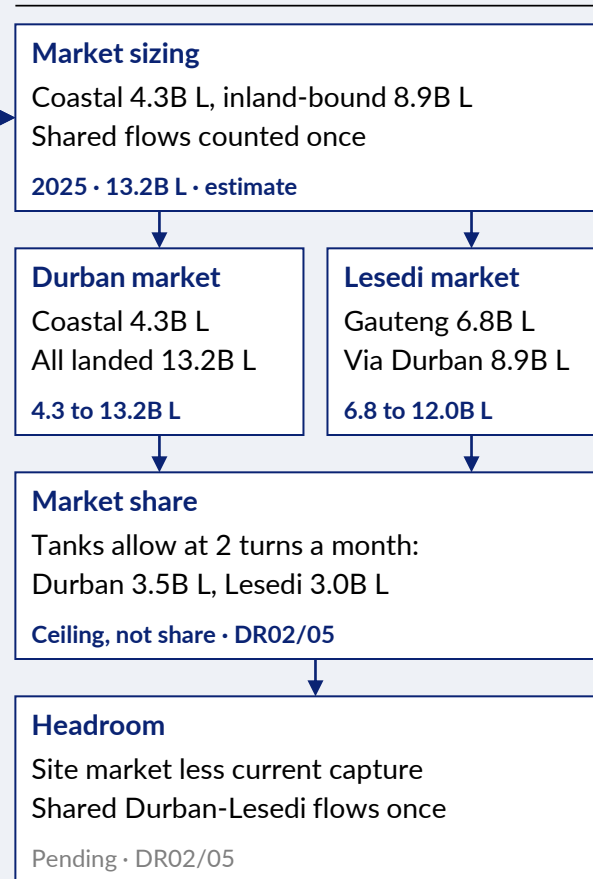
1 NATIONAL DEMAND



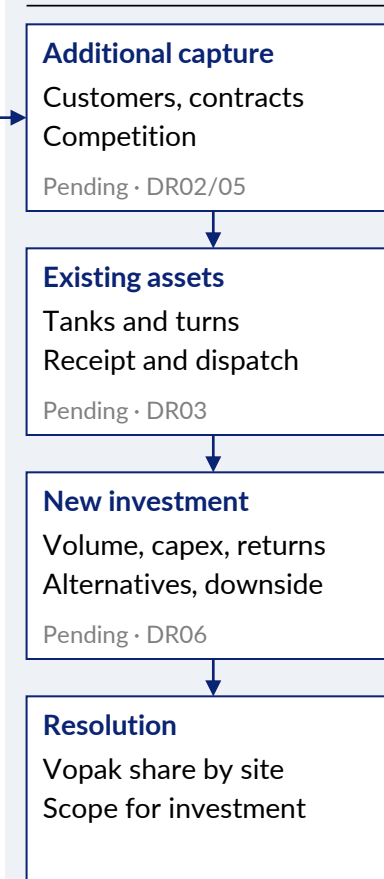
2 SUPPLY AND DELIVERY



3 VOPAK OPPORTUNITY



4 INVESTMENT



Source: 8 Oct PS session; Manish DR01 (sales, SARS customs, energy balance to 2021) and DR04 (entry points); provincial sales 2022. Coastal = KZN, WC, EC. Production is 2021, the last published year. Destinations sum to consumption; routes overlap.

Demand by product and province: where the litres are sold

1 COASTAL • 10.0B L

KwaZulu-Natal • 20%

Petrol 1.5 • diesel 2.8B L
Since 2013: +5%

2022 • 4.3B L • observed

Western Cape • 18%

Petrol 1.4 • diesel 2.5B L
Since 2013: +6%

2022 • 3.9B L • observed

Eastern Cape • 8%

Petrol 0.7 • diesel 1.0B L
Since 2013: +9%

2022 • 1.7B L • observed

2 INLAND • 12.0B L

Gauteng • 31%

Petrol 3.5 • diesel 3.3B L
Since 2013: -2%

2022 • 6.8B L • observed

Free State • 6%

Petrol 0.6 • diesel 0.8B L
Since 2013: -39%

2022 • 1.4B L • observed

Limpopo • 2%

Petrol 0.3 • diesel 0.2B L
Since 2013: -40%

2022 • 0.5B L • observed

Mpumalanga • 8%

Petrol 0.6 • diesel 1.1B L
Since 2013: -24%

2022 • 1.8B L • observed

North West • 5%

Petrol 0.4 • diesel 0.6B L
Since 2013: -16%

2022 • 1.1B L • observed

Northern Cape • 2%

Petrol 0.1 • diesel 0.4B L
Since 2013: -23%

2022 • 0.5B L • observed

3 WHAT IT MEANS

Gauteng anchors inland

6.8B L, 31% of the national total
Lesedi's catchment

2022 • observed • DR01

Diesel grew on the coast

KZN 2.3 to 2.8, WC 2.0 to 2.5B L
Petrol fell in every province

2013-2022 • observed

2023 is estimated

First-quarter shares of the total
Error 2.5 pts petrol, 5.3 diesel

Back-test • DR01

Source: Manish DR01 provincial file: department sales by magisterial district summed to province, 2013-2022 observed; 2023 estimated from first-quarter shares. Coastal = KZN, WC, EC (presentation grouping, not registered). Petrol and diesel.

Demand by use: three sources to calibrate

Source	Road transport	Mining and industry	Agriculture	Power generation	Other and marine
Energy balance 2021, final use	15.1B L Petrol and diesel 2021 • energy balance	1.6B L Incl. mining 1.3 2021 • energy balance	1.2B L Petrol and diesel 2021 • energy balance	Not in final use Transformation input Pending • none published	5.3B L Petrol and diesel 2021 • energy balance
Model baseline 2024, high demand	16.4B L Petrol and diesel 2024 • model, sourced	1.5B L Diesel 2024 • model, sourced	1.1B L Diesel 2024 • model, sourced	3.6B L Diesel 2024 • model, sourced	0.8B L Marine diesel only 2024 • model, sourced
Reported separately	No separate series Inside sales totals Pending • none published	No separate series Inside sales totals Pending • none published	No separate series Inside sales totals Pending • none published	1.6B L FY23 1.3 • FY25 0.9 FY24 peak • DR07	No separate series Inside sales totals Pending • none published

Totals agree, the mix does not

Balance 23.2B L (2021), model 23.4B L (2024)
split across sectors differently

To calibrate

Power is the largest gap

Model 3.6B L against 0.9-1.6B L reported
Eskom plus independents, FY23-FY25

DR07 • to calibrate

"Other" is large in the balance

5.3B L, mostly commercial and public
The model has no matching segment

To calibrate

Source: department energy balance 2021 (petrol and diesel final consumption); Manish sector baselines (model, high demand, 2024; road petrol and diesel, other sectors diesel); DR07 (Eskom reported litres; independents estimated at 0.31 L/kWh). Not reconciled: overlap log item 6.

Demand parameters: the levers and their 2035 cases

1 ECONOMIC ACTIVITY

GDP growth

Baseline: 0.5% a year
2035: 0.7 / 1.6 / 2

Manish revised 3 of 3 • not accepted

Road freight activity

Baseline: 100 (2024 = 100)
2035: 95 / 135 / 150

Manish revised 1 of 3 • not accepted

Real fuel price

Baseline: 100 (2024 = 100)
Petrol 2035: 80 / 100 / 120
Diesel 2035: 80 / 100 / 140

Added by Manish • not accepted

2 VEHICLE FLEET

Battery EV share of new sales

Baseline: 0.2%
2035: 2 / 15 / 40

Manish revised 3 of 3 • not accepted

New-vehicle efficiency

2035 cases, % a year:
Petrol 1 / 1.25 / 1.5
Diesel 0.5 / 0.75 / 1

Nigel proposal kept • not accepted

Hybrid share of new sales

Baseline: 2.1%
2035: 2.1 / 6.7 / 11.3

Added by Manish • not accepted

3 FREIGHT AND RAIL

Road freight moved to rail

Baseline: 0%
2035: 0 / 7 / 13

Manish revised 2 of 3 • not accepted

Electric share of new trucks

Baseline: not reported
2035: 0 / 3 / 9

Added by Manish • not accepted

Rail passenger journeys

Baseline: 74m a year
2035: 100 / 250 / 320

Added by Manish • not accepted

4 POWER

Diesel turbine output

Baseline: 100 (FY24 = 100)
2035: 5 / 15 / 50

Nigel proposal kept • not accepted

Private backup generation

Baseline: not measured
2035: 0 / 0 / 0.5

Added by Manish • not accepted

Reported power diesel

FY24 1.6B L, FY25 0.9B L
Falling as load-shedding ended

Observed and estimated • DR07

Source: Manish lever response (7 Oct): 2035 low / medium / high, an analyst proposal, not accepted; baselines from Stats SA, naamsa, Eskom and SARS. Levers overlap (road activity and rail; EVs and efficiency); the double-count check is open (DR07).

Port and logistics system: how fuel reaches coastal and inland demand

1 PORTS AND LIQUID BULK

Port network

Durban 8 liquid-bulk berths
Richards Bay 2; Island View hub

DR04 • TNPA Our Ports

Fuel imports by port, 2025

Durban 13.2 of 16.7B L (79%)
Cape Town 1.5, Richards Bay 0.7

2025 • petrol, diesel • customs • DR04

All liquids landed, 2025

Durban 20.7 of 31.3M t (66%)
Crude, fuels, gas and chemicals

2025 • TNPA tonnes, all products • DR04

Durban role

Ship receipt; dispatch mainly road
Trunk-line injection to confirm

DR04 • Vopak NERSA filing

2 PIPELINES

Pipeline network

NMPP Durban to Jameson Park
(TM2, 180M L); old DJP retiring

DR04 • Transnet Pipelines 2020

Pipeline throughput

All products FY24 15.2B L
incl. crude and jet; FY25 13.4B L

Pack research • TPL 2024, IR 2025

Capacity and use

148M L/week = ~7.7B L/yr
vs 13.2B L landed at Durban

DR04 • 2020 capacity, lead to check

Lesedi role

4 pipeline intakes, 8 road bays
Customers ship via the trunk line

DR04 • NERSA, Feb 2026

3 RAIL

Rail network

Island View rail gantries
Durban and Matola to Gauteng

Pack research • NERSA 2026

Rail throughput

Durban-Gauteng fuel: not published
Airport jet by rail: 37% to 5%

Pending • Transnet; jet FY18-FY22

Capacity and reliability

Wagon turnaround 33 days
against 14-day design

Pack research • industry, via NERSA

Relevance

Airport jet shifted to pipeline,
rail to pipeline 30:70 to ~10:90

Pack research • NERSA 2026

4 ROAD

Road network

Tankers, coast to inland
Mozambique road imports 0.5%

2025 • customs • DR04

Road throughput

Corridor volume not published
Residual needs a common basis

Pending • DR04 open

Cost and constraints

Pipeline tariff 68 to 77 c/l
Gauteng allowance 83 to 91 c/l

2024-26 • NERSA, dept • DR04

Relevance

Shippers moving to road
Sapref: planned import terminal

DR04, DR08 • news, CEF

Source: Manish DR04, DR08 (8 Oct): SARS customs by office, Transnet Pipelines 2020, NERSA, department zone differentials, CEF. Pack research: Transnet Pipelines 2024, Transnet IR 2025, NERSA jet fuel paper (Jun 2026). Overlaps: overlap log, 8 Oct.

Outstanding data for market share and investment testing

Manish assembles public evidence and estimates. Nigel coordinates client requests. Nigel / Henry review scenario settings.

Outstanding request	Status and what remains open	Coordinator	Decision supported
DR01 National balance	Received 8 Oct: balance reconciled 2014-2021 (customs, sales, energy balance). Open: stock change, production after 2021, 2024-25 sales.	Manish Nigel reviews	Current market size
DR04 Routes and access	Received 8 Oct, partial: entry points, pipeline tariff, zone costs, site access. Open: road and rail rates, pipeline volumes by product.	Manish Nigel reviews	Accessible market at each terminal
DR07 Demand evidence	Received 8 Oct: Eskom litres, plant dates, fleet by province, efficiency, rail. Open: double-count check; sector baseline differs (overlap log).	Manish Nigel reviews	Demand baseline and scenario calibration
DR08 Refinery supply	Received 8 Oct, partial: six plants, capacity and status, Natref split. Open: Secunda and Astron output by product; output after 2021.	Manish Nigel reviews	Domestic output and import requirement
DR02/05 Client capture	36 months of site/product receipts and dispatches, linked transfers, customer destinations and contract volumes.	Nigel coordinates	Current share and achievable capture
DR03 Client operations	Working and committed tanks, turns, occupancy, downtime and receipt/dispatch limits, by site and product.	Nigel coordinates	Usable capacity and bottlenecks
DR06 Investment inputs	Handling/storage terms, operating costs, project capex, timing, tax, working capital and hurdle rate.	Nigel coordinates	Incremental returns and project alternatives

DR09 remains separate: EV adjacency needs customer demand, power access and standalone economics. Nigel coordinates.

Source: DR01-DR09 request register; Manish hand-back of 8 October (DR01, DR04, DR07, DR08; merged as PR #1). Register file not yet updated for receipt. Client requests DR02/05, DR03, DR06: prepared, not sent.

Decisions and next steps

1 DECIDE

Demand by use baseline

Choose or blend the three sources
Start with the power gap

Nigel, Manish · p.4

Demand lever cases

Accept or revise the 2035 cases
Run the double-count check

Nigel, Henry · p.5

Coastal and inland split

Adopt KZN, WC, EC as coastal
Add a register row

Nigel · p.3

2 REQUEST

Client data

Send DR02/05, DR03, DR06
Site throughput, tanks, economics

Nigel · p.7

Operator data

Secunda, Astron output by product
Stock change by product

Nigel · DR08, DR01

Logistics confirmations

Trunk-line capacity today
Durban injection into the NMPP

Nigel · DR04, p.6

3 BUILD NEXT

Market sizing method

Accessible demand by site
Shared Durban-Lesedi flows once

Manish · p.2

Figure check

Extend the pack figure check
to this pack's volume files

Manish · overlap log

Request register

Record DR01, DR04, DR07, DR08
as received on 8 October

Register owner

Owners as proposed in the request register and the 8 October hand-back; no dates are committed. Page references are to this pack; open items are listed in workstreams/WSO_governance/workplan/overlap_log_2026_10_08.md.